

General Information

Communication Channels

- [InsertEmail](#) - any type of appointments, e.g., booking the lab for testing or pilot experiments
- [InsertEmail](#) - Official appointments only, linked to the participant database
- 'EEG Lab 8 (A134)' Mattermost Channel: day-to-day communication, joint by anyone working in lab

Access to the Lab

- Reception → (key #XX / EEG8 / A134) # they will ask you to sign on key control logs

Lab Journal

- Document incidents during your experimental session, at minimum the following:
your name + study name (respirationAdj1/rsAirflow), which batteries (for amplifiers) used, which cap used.

Any consumables (soon) running out?

- e.g., gel, syringes, alcohol swabs, wooden sticks, kitchen rolls, towels ...
- → write into 'EEG Lab 8 (A134)'! # we all need it ☺

Still enough money in Handkasse?

- Not needed, everything via bank transfer

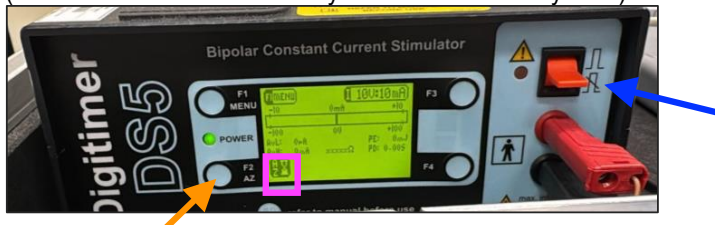
Other relevant links

- Manuals, Certificates, Notes: [InsertLink](#)
- Lab Manual: [InsertLink](#)



Trouble Shooting (!)

- Check Digitimer (DS5) interface regularly! **We ALWAYS want this [Pink] to go away / best not to show up** (can mean that stimulation may not be delivered anymore):



Between experimental blocks (breaks), always do the following:

- Press red switch (top left, **Blue**) downward.
 - Press button (black circle at bottom left, **Orange**) labeled "F2 AZ"
 - Push red switch (top left, **Blue**) upward.
 - **Case ID from SosciSurvey missing?** → text Lioba!
 - You **cannot log into the EEG Laptop** (right monitor on desk)? → Did you select **Windows**?
 - **No Finalization of Online Survey in time?**
 - Send your own reminder email to participant, as soon as you notice.
 - Check whether there is enough time (50 min extra!) to prolong laboratory session
EEG8 free long enough?
- If not:** Cancel session + postpone / try to find a new date!
-

Preparation after Scheduling + Before Session

- **Check your cbs email address:** you should have received (1) notification on digital informed consent signature, and (2) notification Case ID has finished base/follow-up survey.
check process: '/GWDG_Owncloud/.../multi_session/organisation/soscisurvey/soscisurvey_schematic.pdf'
 - **Prepare:** Informed Consent (in digital form + signed by you [experimenter] digitally) & send out
 - **Prepare:** Case ID (random number from SosciSurvey)
Where find: sosci.cbs.mpg.de → LogIn: enk@cbs.mpg.de PW: [askLioba](#) (2-Factor-Authentication)
[1] Project: nro-consent → Collected Data (left side bar) → View Dataset
→ IC02_01 + IC02_02: FirstName LastName, REF = Case ID
[2] Project: nro-259 → Collected Data (left side bar) → View Dataset
→ REF = Case ID
- **A134, drawer: Nose clip, d2-Revision (1x Kurzanleitung, 1x Testbogen)**

Preparation (30 min before session starts)

- Switch on...
 - EEG laptop: ... / ... # **WINDOWS!** # stores data; *note*, left monitor on desk is for laptop!
 - Gullveig: ... / ... # **WINDOWS!**
 - Black Boxes on Equipment Shelf Tower (top: recording, bottom: synch box) # need time!
 - Digitimer DS5 (D2)
 - National Instruments (NI USB-6343) Box
- Check connections:
 - Is LPT (from laptop) connected to Gullveig's COM2 entry (next to tape) ? # should be the case (for exp_condition triggers) # only this one of Gullveig's entries works: LPT2, address: hex2dec('D010')
 - Is 'A In' cable connected to DS5? # should be the case (for stimulus triggers)
- Open NeurOne (Recording) Software (left monitor on desk)
 - → Tab **Protocol**
 - Our protocol: **62 EEG_ECG_Resp_Lioba_rsAirflow** → Select
 - → Tab **Person**
 - → Click 'New' (Green top left) → Label (**Person ID**):
sub0XX_RespAdj1
Example: sub001_RespAdj1 (Example for Participant ID 1)
- Select batteries for amplifiers:
 - Old Battery (#2): 6V charger (labelled OLD)
 - New Battery (#3): 7.2V charger (labelled NEW)
 - **Go:** NeurOne Recording Software Interface → Our Protocol → Plug-Ins → Voltage Monitor:
Select the value that matches the voltage of your battery (6V vs. 7.2V)
- Open Experimental Code (monitor in middle on desk):
 - Our Code: [InsertLink](#)
- Prepare breathing control accessoires:
 - **Nose clip**
 - Mouth Tape
- Prepare **d2-Revision paper-pencil assessment: Kurzanleitung + Testbogen**
- Arrange necessities for EEG preparation:
 - Clean cup/glass, Water (for participant)
 - Tape measure (head circumference), towel (for shoulders), felt pen (for Cz position),
2 alcohol swabs (for EOG & ECG placement),
 - 2 cotton swabs + tiny alcohol bottle
 - 4 syringes filled with gel (default: blue lid, allergic/sensitive skin: white lid),
1 syringe filled with fluid gel (medium size, red lid), all with blunted needles
 - 2 single electrode strings (for EOG & ECG placement) # only use **RED ONES**
 - Respiration Thermistor
- Prepare setting in EEG cabine:
 - Monitor: Set to **S = D-SUB / B**
 - S (2nd to left button) = DVI → Gullveig
= D-SUB → switch...
 - A → Torsten
 - B → EEG laptop (to show signal / impedance test)

Preparation the Participant + Recordings

- When reception calls, pick up participant and bring to EEG8 / A134.
- **Inform them once again about the session:** (Online Questionnaires done already), **now:** 1 more online quest, 1 paper-pencil test, EEG cap preparation, Resting State measure (nose/mouth), Experimental Task.
 - **Tell them:** Informed Consent was signed online, will be send via e-mail to them!
- **Bathroom needed? (Recommend to them)**

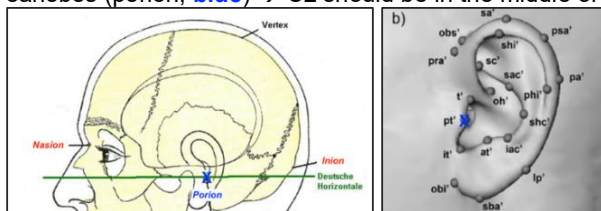
- **STAI-state** # you need their **SosciSurvey Case ID!**

Link: [InsertLink](#)

- **Paper-pencil exercise (d2-Revision):**
 - **Kurzanleitung** # let them do the 2 exercises on the sheet
 - Get **Testbogen + Pen** ready # **don't show** before you are ready
 - **Your stopwatch** (on your phone!): for every single line (14 lines! So concentrate, please!): **20 seconds per line sharp: 'Halt, Nächste Zeile!'**

• EEG Preparation

- Measure circumference
- Put on cap
 - For better precision: measure the distance between nasion and inion [red] as well as between earlobes (porion, blue) → **Cz** should be in the middle of both measurements

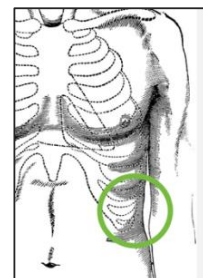
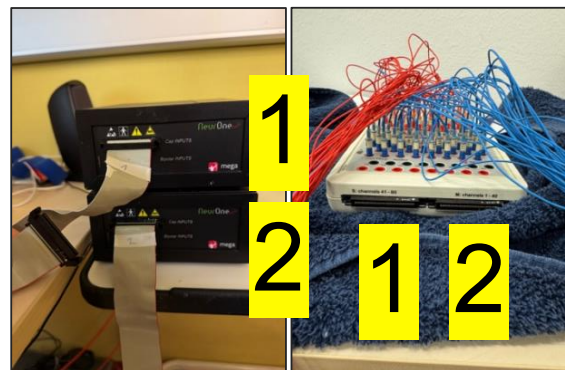


- **Electrode Digitization** (approx 10 min)

- Guide to **EEG7**
- See Appendix for **Digitization Protocol**

- Guide participant into EEG cabine
- Put towel over the shoulders
- **Connect EEG cable strings to amplifiers**
- **NeurOne (Recording) Software:**
 - Tab **Measurement**
 - Click: Impedance Test # check, can you see impedance display in cabine? (switch monitor)
- Procedure

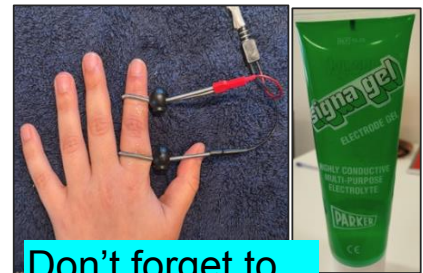
- Alcohol with cotton swaps – recirculate in each electrode hole once
- Abrasive (sandy) gel (blue/white lid)
 - [1] switch ground and reference electrode at the start (need low impedance) # keep in mind: you need to restart impedance test (!) When **<1 kOhm** → switch back
 - [2] attach ECG electrode + EOG electrode (single electrode strings):
 - **EOG: 1 cm below right eye:** clean with alcohol swap
 - **ECG: below the left breast [green circle]**
 - clean both positions with alcohol swap
 - attach electrode with sticker
 - same procedure: bring down impedance with sandy gel, then add fluid gel
 - tape over position to make it stick



- [2] reduce impedance until all green (that is, **<5 kOhm**)
takes ages, keep participant entertained ☺ (40-60 min)
- Fluid gel (medium sized, red lid) – insert in each electrode hole once only
- **Attach respiration thermistor**
- Click on Start (Measurement 'Session' starts & red text appears: **'Not recording'**) and **explain to participant:** (Note regarding data storage: session folder is now created)
 - Blinking (therefore our EOG), Swallowing (try not to!), Chewing (No gum! try not to!), Teeth Clenching (try not to!), Yawning (try not to!), Head Moving (try not to!)

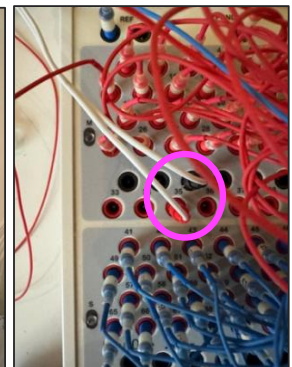
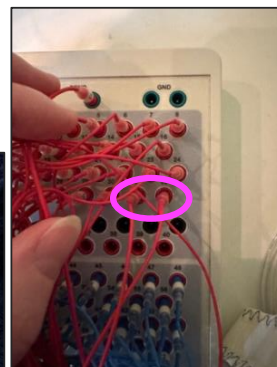
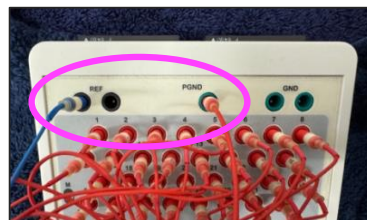
Check in NeurOne (Recording) Software (top!):
Is Protocol correct? Is Person ID correct?!?!?!?

- Ring electrodes to Left Index Finger
 - [1] **Attach: Red Top, Black Bottom**
 - [2] Surround with **Gel (Green Tube)**
 - [3] Place **kitchen roll** below hand & between Index + Middle
 - # make sure, index and middle finger don't touch! Inform participant about importance of this!
 - [4] Palm of hand should face downwards!
 - [5] Place pillow below left elbow, ask participant to relax arm



Don't forget to separate fingers with kitchen roll!

- **Final configuration of the NeurOne JackBox:**
2 (blue + red) braids of electrodes for EEG recordings
FCz = **REF**, POz = **Ground (PGND)**
35 (bipolar: red + black) = **Resp**
31 = **ECG**, 32 = **EOG**



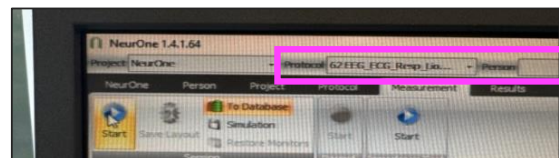
- Have another look at all the channels in the recordings: Are you happy? ☺

Experimental Session

Check battery power from time to time! Change to a fully charged one, if the battery is low!
Add notes to rsAirflow Handbook (Any weird electrodes? Any noisy recording due to teeth clenching/else?
Any problems? Any strange threshold assessment? ...)

Resting State

- Explain (Nose clip vs. Mouth Tape)
- **Check in NeurOne (Recording) Software (top!):**
Is Protocol correct? Is Person ID correct?!?!?!?



Start recording

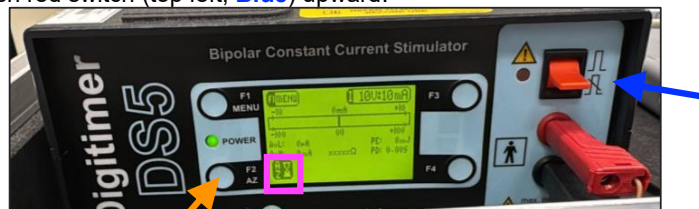
(Note regarding data storage: Recording File ('Phase' in NeurOne) is opened within Session Folder now!)

Red Text (**'Not recording'**) should not be there anymore!

- **Start Resting State Script**
- Once done, **Stop EEG recording.**

Experiment

- **Script RespirationAdj1:**
- Participant Information needed:
 - Participant No. (1-50)
 - Age (years, 18-35)
 - Gender (f,m,d)
 - EEG cap, cm (e.g., 56)
 - SosciSurvey Case ID **# needs to be prepared before session**
- Procedure
 - [1] Let participant feel electrical stimulus at finger a couple of times
Some people have experienced an electrical stimulus (e.g., from electric fence for cattle), others have not. For you to know how it feels, we will deliver supra-threshold stimuli. The stimuli in experiment will be much lower, barely noticeable for you.
 - [2]
Tape over mouth! (Nose breathing only)
Threshold (Thr1F) Assessment # script is automatic, however you [experimenter] need to set final threshold mA intensity for the experimental block. Sometimes it is advised to repeat the assessment.
 - Example Case 1: Psychometric curve is too steep (almost 90° / vertical)
 - Example Case 2: Participant performs very poorly on last trials (at end of Thr1F)
 - Example Case 3: Outcome mA intensity is very far from previous intensities ($\sim \pm 1$ mA)
 - [3]
– **Tape over mouth! (Nose breathing only)**
– **Start EEG recording # Red Text ('Not recording') should not be there anymore!**
Experimental Block
 - Remind before: No yawning, moving, teeth clenching, chewing, etc. **please!** ☺
 - **After every block (!!!!) :**
– **Stop EEG recording** (Note on data storage, 'Phase' is stored)
– **Always do the following:**
 - Press red switch (top left, **Blue**) downward.
 - Press button (black circle at bottom left, **Orange**) labeled "F2 AZ"
 - Push red switch (top left, **Blue**) upward.



- Why? We don't want the symbol in **Pink!**

- **Breaks:**
 - Water for participant? **add gel to finger electrodes (after block 3 & 6)?** Motivate participant: experiment so important to you/team, you know how horrible psychophysical experiments can be, empathize, be kind.
 - **DON'T explain the purpose of the experiment!** If participant asks, tell them you cannot inform them, participants need to be naïve to study purpose and should simply do their best.

Equipment Removal

- Take of cap & put it in **green small bucket** to soak

Guide Participant to Hair Cleaning

- Hand out **Participant entrance card** # needed to return to laboratory corridor / A134
- Guide to **hair-washing room (2nd floor)**. Fresh towels are on small shelf, used towels go into bin.

Compensation

- Participants receive (in cash):

8.50 €	# fixed amount for online questionnaires
15.00 € / hour	# for laboratory session (including EEG)
- Examples:
 - 4 hours in Lab + 8.50 € → 68.50 €
 - 4 hours 15 min in Lab + 8.50 € → 72.25 €
 - 4 hours 30 min in Lab + 8.50 € → 76.00 €
 - 4 hours 45 min in Lab + 8.50 € → 79.75 €
 - 5 hours in Lab + 8.50 € → 83.50 €
 - 5 hours 15 min in Lab + 8.50 € → 87.25 €
 - 5 hours 30 min in Lab + 8.50 € → 91.00 €
- Guide down to reception/exit. Thank you & Good bye.

Charging EEG batteries

Fully charged batteries last 4 hours, that is why you should never forget charging them.

- Which batteries** did you use? → write into Lab Manual
- Use the **correct charger** for your batteries (labelled):
 - No. 2 (old): use OLD charger
 - No. 3 (new): use NEW charger
 - No. 4 (new) # runs out quickly, so try to avoid
- Check charging status: [1] OLD charger: blinking green (charging), no light (done).
[2] NEW charger: blinking red (charging), green (done).

Data Transfer & Storage

It can take time! Be careful and concentrated!

- Raw EEG data stored in **E:/...NOBACKUP**
- data/** InsertFolderName /Study1_respirationAdj1/
 - Don't be surprised when you see more than n_blocks files! Be aware of this difference:
 - 'Session' for NeurOne: When you click Start (e.g., to show participant recording)
 - 'Phase' (within a session) for NeurOne: When you click Start (within the Start Session, i.e., after you have already clicked Start once).

**Transfer
digitization file?
use this USB**



EEG Equipment Cleaning

*Always, always, always keep **JackBox** wrapped in towel and away from water!*

Leave the Lab

*Everything clean? **EEG Batteries** charging? Wrote into **Lab Manual**? Return Lab Keys to reception!*